A Preference-Driven Malicious Platform Detection Mechanism for Users in Mobile Crowdsensing

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***Abstract*— Exploiting mobile crowdsensing to conduct data collection and analysis brings unprecedented opportunities to promote the development of the Internet of Things(IoT). However, malicious platforms may provide untrusted data or illegally leak users' information, which leads users in crowdsensing networks to be reluctant to participate in sensing activities. Besides, users are unwilling to report malicious platforms without sufficient incentives. To tackle the problem, a new incentive mechanism is proposed by modeling users' preferences in this paper. Specifically, two scenarios are considered to detect malicious platforms when users join sensing activities according to the system grasps user's information, i.e., complete information scenario and partial information scenario. Different incentive algorithms are designed for each scenario to optimize the systems incentive cost. In the complete information scenario, we minimize the total incentive cost by ranking users' preferences. In the partial information scenario, uniform Distribution and Laplace Distribution are employed to model the distribution of users' preferences to find the optimal cost. Specifically, we incorporate the concept of non-convexity into design the incentive mechanism, when user preferences obey the Laplace Distribution. By conducting an in-depth exploration the properties of Laplace Distribution, we can transform it into a convex problem to solve it efficiently. The analysis based on these mechanisms lays a theoretical foundation on the detection of malicious platforms. Furthermore, the soundness of modeling and the accuracy of analysis are verified through extensive simulation, which also guides the design of more sophisticated incentive schemes for the detection of malicious platforms.**

***Index Terms*— Mobile crowdsensing, malicious platforms, incentive mechanism, uniform distribution, Laplace distribution.**

I. INTRODUCTION

MOBILE CrowdSensing (MCS) is a new data acquisition model that combines crowdsourcing ideas and mobile device sensing capabilities. The research shows that MCS has the advantages of flexible deployment [1], low requirements on users knowledge [2], low deployment cost [3], etc. Therefore, adopting MCS to carry out data collection and analysis has become a mainstream hotspot. Thus far, MCS has been widely used in various fields, e.g., environmental monitoring, smart cities and intelligent transportation systems (ITS) [4], [5], [6].

A typical system framework for MCS is shown in Fig. 1, which includes three parts: the sensing platform, sensing users and task providers [7]. After receiving service requests from the task provider, the sensing platform assigns sensing tasks [8], [9], [10], [11]. to sensing users. Meanwhile, the sensing platform also has the ability to handle sensing data and perform other management functions. After receiving the sensing task, sensing users will perform the collection of sensing data [12], [13], [14]. Then, sensing users upload the data to the sensing platform [15]. When the sensing platform returns the sensing data to the task provider, a complete sensing link is completed. It is noticed that compared to traditional wireless sensor networks (WSNs) [16], there is a significant advantage of MCS in collecting sensing data.

Although MCS emerges as the significant functionality for promotion sensing services, new issues and challenges are also exposed due to the following factors. On one hand, malicious sensing platforms may provide false information or expose users' privacy [17], [18]. for illegal profits, which may destroy the security of MCS and users' reluctance to actively participate in sensing tasks. On the other hand, if there is not sufficient incentive, users are reluctant to actively report malicious sensing platforms in MCS, which will reduce the number of users participating in sensing tasks. Therefore, designing a reliable incentive scheme in MCS to help the system detect malicious sensing platforms remains an open and vital problem.

In mobile crowdsensing, since the difference between normal platforms and malicious platforms is not obvious, it is difficult for the sensing system to directly detect malicious platforms by the historical behavior of platforms. Moreover, the detection of malicious platforms can take more time and cost, when the system is directly involved. Therefore, in this paper, a new approach is proposed from a fresh perspective, i.e., a user-incentive-based detecting mechanism for malicious sensing platforms.

However, user preferences can usually lead to users having insufficient incentives to report malicious platforms. In this paper, user preference is defined as the rational and tendentious choice made by the user in the face of malicious

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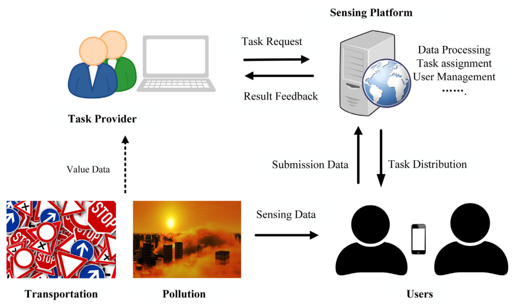

Fig. 1. A typical architecture of a crowd sensing system.

platforms. Specifically, user preference in this paper is similar to human preference, which is actually the result of a combination of multiple factors. In MCS, users have self-will and exhibit varied preferences, which significantly has cast on their responses and attitudes to malicious platforms, e.g., some users are more tolerant of ad injections from malicious platforms. Furthermore, in order to avoid being reported by users, malicious platforms may offer compensation, such as coupons and money. Therefore, we analyze the user preferences and the possible compensation offered by malicious platforms. We combine Continuous Probability Distribution [19] with MCS to stimulate users to report malicious sensing platforms. The basic idea is to establish a user reporting scheme based on the interaction behaviours between users and sensing platforms, where the malicious platforms will be restricted or eliminated for MCS to enhance the service security. In this method, the system administrator is not directly involved in the detection process. Instead, the approach achieves malicious platform screening by incentivizing users with detection tasks. When the number of times a sensing platform has been reported exceeds the threshold we have set, the system will determine if it is a malicious platform. Meanwhile, the designed method under Continuous Probability Distribution can guarantee the individual rationality (IR) [20] and incentive compatibility (IC) [21] properties.

The major contributions of this paper are threefold.

- We propose a novel and efficient method, i.e., crowdsensing, to detect malicious platforms in sensing network. Moreover, to motivate enough users to participate in the detection task, we design the corresponding user incentive model.
- We consider two scenarios, i.e., the complete information scenario and the partial information scenario, and design the incentive mechanism separately. In the complete information scenario, optimal incentives are designed based on the analysis of user preferences. In the partial information scenario, we take the premise that users' preferences conform to the Uniform Distribution and the Laplace Distribution while developing the corresponding incentives to optimize the system cost.
- Besides the proof and analysis of the correctness and complexity, we conduct extensive simulations to evaluate the performance of our scheme. The numerical results show that the proposed scheme can effectively improve the quality of sensing service and minimize the system incentive costs.

The rest of this paper is organized as following: Section II reviews the related work. In section III, we introduce the problem description and the system model. The model of malicious platform detection based on continuous type distribution is described in section IV. Performance evaluations are shown in section V, and section VI concludes this paper.

II. Related Work

In this section, we review the related work including the trust evaluation scheme, the threshold-based filtering scheme and the classification-based filtering scheme.

A. The Trust-Based Evaluation Scheme

Trust evaluation is an important factor to evaluate the credibility of users and sensing platforms in MCS. Xia et al. [22] proposed a novel trust inference model, where two trust attributes named subjective trust and recommendation trust, are selected to quantify the trust level of specific nodes. The authors combine subjective trust and recommendation trust to obtain the overall trust of the nodes, which can be used to classify nodes as either honest or malevolent. Liu et al. [23] designed a novel consensus mechanism to guarantee decentralization, i.e., a decentralized intrusion detection framework for network trust evaluation, where intrusion detection and network trust evaluation are integrated into node detection. Since genuine information is discrete, the literature [24] showed a technique for computing malicious node impact for a dynamic VANET topology. The author develop a twofold framework to change logical information to continuous information, which analyzes to deal with the correlation between node discovery of exchanged information and impact of malicious nodes, to assign score to each malicious data for the identification of malicious vehicular nodes. Hu et al. [25] proposed a reliable trust-based recommendation scheme for platooning services(REPLACE), which collects and models feedback from users to design the reputation system. Liu et al. [26] proposed an approach called Perceptron Detection (PD), which uses both perceptron and K-means methods to compute IoT nodes' trust values and detect malicious nodes accordingly.

B. The Threshold-Based Filtering Scheme

To solve the problem that high thresholds fail to detect malicious nodes and low thresholds may increase the false alarm rate, an outlier-based adaptive threshold approach is proposed by Chae et al. [27]. The authors created effective rules to analyze internet packets collected from dynamic environments based on historical information. The result shows that high thresholds reduce the true positive rate and low thresholds increase the false positive rate. Meng et al. [28] designed a blockchain-based trust management scheme, which dynamically set thresholds to identify malicious nodes more efficiently. Liu et al. [29] proposed a malicious model detection mechanism based on the isolation forest (iforest), named D2MIF. In the D2MIF, an iforest is constructed to compute the malicious score for each model uploaded by the corresponding participant, and then, the models will be filtered if their malicious scores are higher than the threshold, which is dynamically adjusted using reinforcement learning (RL). In the [30], Liu et al. proposed a simple voting scheme to identify malicious nodes, which uses the 1/0 decision. It considers nodes with more than half of the votes as malicious, however, this scheme is not suitable for the scenario where the number of nodes is small.

C. The Classification-Based Filtering Scheme

Singh et al. [31] proposed a framework for the detection of malicious users, non-malicious users and celebrities, which uses an attribute set for user classification based on user characteristics. The authors analyzed the detection mechanism used for spammers and calculated two parameters user score and tweet score to detect malicious users. In the [32], a malicious detection method based on multi-variate classification is proposed by Dai. To filter malicious nodes, the author extracted the preferences with known nodes to generate classifiers. Abubaker et al. [33] utilized federated learning for the detection of malicious nodes, where Support Vector Machine (SVM) and Random Forest (RF) are used in federated learning to identify and remove malicious nodes.

In the trust-based evaluation scenario, additional costs will be incurred when the system evaluates the trust of the users. Some malicious users cannot be recognized by the system if a reasonable threshold cannot be set in the threshold-based filtering scenario. It is also a challenge to ensure fairness when using voting to screen malicious users. Most of these existing efforts are not applicable to the scenario, where the system cannot easily detect malicious users directly. Moreover, when malicious users provide motivation to other normal users, the normal user may show dishonesty. Therefore, we propose a novel malicious user detection scheme by modeling the preference distribution of users.

III. Problem Description and System Model

Currently, most of the current motivational work investigates the scenario, where a single sensing platform releases sensing tasks [15]. Nevertheless, users usually receive sensing tasks from different platforms at the same time. Therefore, the user's choice of sensing platforms is crucial. In general, voluntary participation in the crowd sensing platform may not be sustainable. This is from the fact that mobile users need to spend their own resources, e.g., smart phone battery, CPU computing power, storage memory, to accomplish the sensing tasks [34], [35]. In order to save energy and maximize profitability, users will purposefully choose the sensing platform. However, malicious sensing platforms can launch ad injections, phishing attacks and expose users' privacy. Thus it is particularly important to design a reasonable mechanism to help the system detect malicious sensing platforms and motivate users to participate in tasks.

As shown in Fig. 2, we consider MCS networks consisting of a number of mobile users and sensing platforms. When the network size is relatively large, it is difficult for the system to detect sensing malicious platforms independently. Therefore, the system can indirectly detect malicious sensing platforms by analyzing the historical behavior of some users and relying on reports from these users. Depending on the characteristic of the sensing network, we assume that users are independent of each other in the system. Thus, the system must provide certain inducements which motivate the user to report MSP. Then, we consider a set of users denoted by $u = \{u_1, u_2 \cdots u_n\}$, where u_i is able to choose the sensing platform by their own wish.

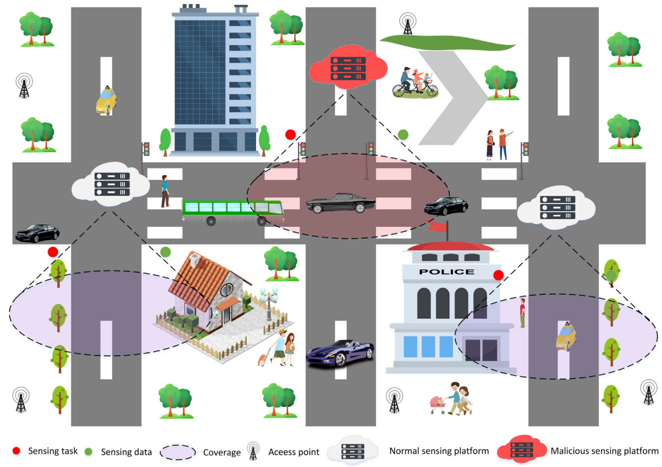

Fig. 2. Framework of MCS networks.

For convenience, we classify sensing platforms into two types, one called NSP(normal sensing platform) and the other called MSP(malicious sensing platform). To encourage users to join sensing tasks, the NSP must compensate users for the cost of performing sensing tasks. Moreover, the MSP also offers certain motivation.[1] In general, the motivation is related to users preferences p_i. When the motivation provided by the MSP meets exactly the u_i's preference, the user's favorable opinion of the platform is positive, and the opposite is negative. This leads to the fact that the MSP may have different motivations for users with different preferences.

For simplicity, we define the motivation provided by the MSP as S, where the S is a constant. After users understand the behavior of the MSP and systems, users can choose whether to report the MSP. We set the utility for the u_i to report and not report MSP as m_i and $S + p_i$, respectively. Thus, the utility function of u_i can be defined as $U_i = max\{m_i, S + p_i\}$. In the system, C_i is used to indicate the u_i's choice. Assuming that C_i obeys the Bernoulli Distribution [36], which can be expressed as

$$C_i = \begin{cases} 1 & (S + p_i < m_i) \\ 0 & (S + p_i > m_i), \end{cases} \tag{1}$$

Here, 1 and 0 indicate that the u_i report and not report the MSP, respectively. Clearly, the cost of the incentive provided by the system is $M = \sum_{i=1}^{N} m_i$.

We stipulate that the system rewards reporting users when more than N_0 users report the MSP. N_0 is our predefined threshold and the model's criterion for choosing N_0 is to reduce the likelihood that users may arbitrarily adopt

[1] Motivation, The MSP offers benefits to users in order to avoid being reported.

'ballot stuffing' or 'bad-mouthing' methods to gain benefits. In general, if users adopt 'ballot stuffing' or 'bad-mouthing' methods, then the number of users who choose to report MSP is difficult to exceed N_0. Therefore, they can't actually get additional benefits.

To avoid malicious users from providing false feedback to discredit the platform, we propose two approaches, 'appeal' and 'punishment'. 'Appeal' means that the reported perceived platform can provide feedback to the system, and the system will determine whether the platform is malicious based on its historical behavior. 'Punishment' means that the user who reports the sensing platform will be flagged once the sensing platform has been determined by the system to be non-malicious. When the number of times a user has been flagged exceeds the set threshold, the system will limit the user to a period of time where the user will not be able to participate in the activities of the reporting sensing platform and any sensing tasks to earn rewards.

To reduce the incentive cost of the system, it needn't to provide additional incentives to users other than N_0. For the MSP, since the preference of the user cannot be known, it will provide the same motivation to each user. Specifically, the problem investigated in this paper is how to ensure that more than N_0 users report the MSP and minimize the system's cost.

IV. Incentive Mechanism Design

In this paper, we divide the system's knowledge of user's preferences into two types: complete information scenario and partial information scenario. In the complete information scenario, the system has a clear understanding of the user's preferences. However, in the partial information scenario, the user's historical behavior is the only way for the system to understand the user's preferences. This section focuses on these two scenarios to design an incentive mechanism that minimizes the incentive cost of the system while screening for malicious platforms.

A. Complete Information Scenario

Focusing on the complete scenario, where the users' information are shared, the system can get the preference of each user. Then, sorted result on the users' collection preferences is obtained, which guide the system's decision. For $i < N_0$, we discuss different cases.

Case1: if $S + p_i \leq 0$, then the u_i has sufficient motivation to report the MSP, so the system can set the user's incentive to 0.

Case2: if $S + p_i > 0$, then the system can set the u_i's incentive to $S + p_i$.

On the other side, the system can also categorize users based on preferences. For the scenario, where users are categorized, users under the same category have the same preferences. However, this analysis of this situation is the same as the ranking of preferences.

B. Partial Information Scenario

Targeting on the partial information, where users are reluctant to share their preferences, the system can only accumulate their preferences by observing users' interactions over a long period of time, i.e., the partial information is carried out in on-demand manner. In this scenario, the incentive mechanism we designed satisfies

$$\sum_{i=1}^{N} P[C_i = 1] = \sum_{i=1}^{N} P[m_i \geq S + p_i] \geq N_0, \tag{2}$$

where $P[C_i = 1]$ and $P[m_i \geq s_i + p_i]$ represent the probabilities of $C_i = 1$ and $m_i \geq s_i + p_i$, respectively. Not surprisingly, the incentive cost of the system increases with $P[C_i = 1]$ and $P[m_i \geq s_i + p_i]$. Thus when the incentive cost is minimal, we have

$$\sum_{i=1}^{N} P[C_i = 1] = \sum_{i=1}^{N} P[m_i \geq S + p_i] = N_0. \tag{3}$$

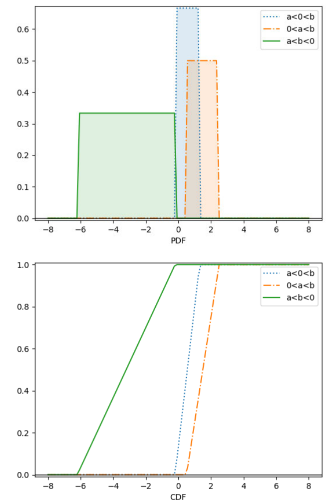

Fig. 3. The PDF and CDF of uniform distribution.

1) Uniform Distribution: When the system only knows the range of values of user preferences, but not the specific preferences of each user, Literature [37] consider the values of user preferences within the range of values to be equiprobable and user preferences conform to the Uniform Distribution. Therefore, we also assume that users' preferences conform to the Uniform Distribution in this paper. Then, the probability density function(PDF) of $S + p_i$ is set to

$$X \sim Uniform(a, b),$$
$$f(x) = \begin{cases} 0 & x < a \\ \dfrac{1}{b-a} & a \leq x \leq b \\ 0 & x > b. \end{cases} \tag{4}$$

The cumulative distribution function(CDF) of $S + p_i$ can be expressed as

$$F(x) = \begin{cases} \int_{-\infty}^{a} 0dx, & x < a \\ \int_{-\infty}^{a} 0dx + \int_{a}^{x} \dfrac{1}{b-a}dx, & a \leq x \leq b \\ \int_{-\infty}^{a} 0dx + \int_{a}^{x} \dfrac{1}{b-a}dx + \int_{x}^{+\infty} 0dx, & x > b. \end{cases} \tag{5}$$

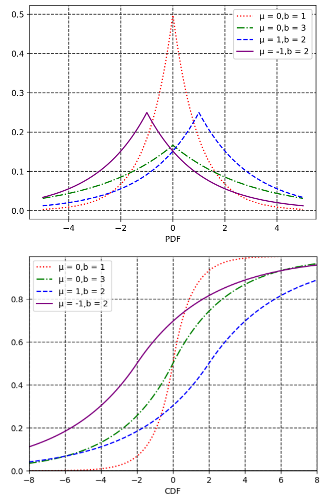

Fig. 4. The PDF and CDF of laplace distribution.

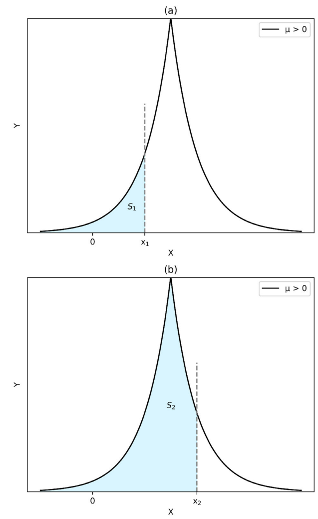

Fig. 5. The shaded area under different values.

where a and b are the lower and upper limits of the Uniform Distribution, respectively. For different values of a and b, the images of CDF and PDF are shown in Fig. 3.

We only consider the scenario, where $a \le x \le b$. If $b < 0$, we can see that $S + p_i < 0$, which means that the users in the system have enough motivation to report the MSP. Therefore, the system does not provide any rewards. When $a \le 0$ and $b > 0$, we discuss two cases.

Case1: if $F(x) = N \int_a^0 \frac{1}{b-a} dx < N_0$, we have

$$F(x) = \sum_{i=1}^{N} \int_0^{m_i} \frac{1}{b-a} dx = \sum_{i=1}^{N} \frac{m_i - a}{b-a} = N_0, \tag{6}$$

from which we know that

$$\begin{aligned} M &= \sum_{i=1}^{N} m_i \\ &= \frac{1}{b-a} \sum_{i=1}^{N} m_i - N \frac{a}{b-a} \\ &= (b-a)N_0 + Na, \quad a \le m_i \le b. \end{aligned} \tag{7}$$

Hence, the system can set the rewards for each user to $\frac{(b-a)N_0}{N} + a$.

Case2: if $F(x) = N \int_a^0 \frac{1}{b-a} dx \ge N_0$, we have

$$N \int_0^b \frac{1}{b-a} dx = 1 - N \int_a^0 \frac{1}{b-a} dx, \quad a \le m_i \le b. \tag{8}$$

Then,

$$N \int_0^b \frac{1}{b-a} dx \le N_0. \tag{9}$$

In this case, the system does not provide any rewards for users according to Eq.(2).

We assume that there are N_0' users, who are motivated by the system at $a > 0$. It is obvious that the range of values of N_0' is between 0 and N. Then, we can notice that the value of m_i is zero when $m_i < a$. Therefore, we have

$$\sum_{i=1}^{N_0'} \int_0^{m_i} \frac{1}{b-a} dx = N_0. \tag{10}$$

Then, the total cost of the system can be expressed as

$$\begin{aligned} M &= \sum_{i=1}^{N_0'} m_i \\ &= \frac{1}{b-a} \sum_{i=1}^{N_0'} m_i - N \frac{a}{b-a} \\ &= (b-a)N_0 + N_0' a, \quad a \le m_i \le b. \end{aligned} \tag{11}$$

Based on the Eq.(11), the total cost of the system has a declining trend with the m_i increase. Due to $a \le m_i \le b$, we can get

$$\begin{aligned} &N_0' a \le \sum_{i=1}^{N_0'} m_i \le N_0' b, \\ &N_0' a \le (b-a)N_0 + N_0' a \le N_0' b, \end{aligned} \tag{12}$$

from which the $N_0' \ge N_0$ can be obtained. It has been theoretically proved that a necessary and sufficient condition for $N_0' = N_0$ is $M = N_0 b$. Therefore, b is the motivation provided by the system to the N_0 users in the system.

2) Laplace Distribution: In general, user preferences are influenced by several factors, e.g., user attributes (age, hobbies, and temperament), the state of the device, and the degree to which the sensing platform affects the user, each of which is limited in its effect. Moreover, user preferences may differ at different times and are independent. Laplace Distribution [38] has the properties of symmetry, sharp peaks and thick tails, and adjustable parameters which make it flexible to better describe the range of values and degree of user preferences. Therefore, according to the law of large numbers, we adopt the Laplace Distribution to model the users' preferences. This means that we can denote the PDF of $S + p_i$ as

$$\begin{aligned} &X \sim Laplace(\mu, b), \\ &f(x|\mu, b) = \frac{1}{2b} exp(-\frac{|x-\mu|}{b}), \end{aligned} \tag{13}$$

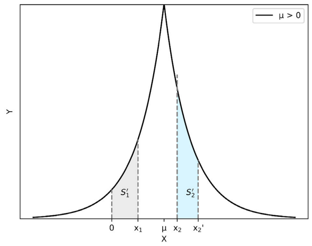

Fig. 6. x_1 moves to 0.

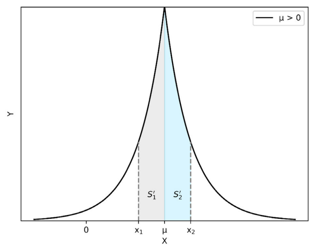

Fig. 7. x_1 and x_2 move to μ.

where μ is the position parameter and b is the scale parameter. Then, the CDF of Laplace Distribution can be expressed as

$$\begin{aligned} F(x) &= \int_{-\infty}^{x} f(u)du \\ &= \begin{cases} 1 - \dfrac{1}{2}exp(-\dfrac{x-\mu}{b}), & x \geq \mu \\ \dfrac{1}{2}exp(-\dfrac{\mu - x}{b}), & x < \mu \end{cases} \\ &= \frac{1}{2}[1 + sgn(x-\mu)(1 - exp(-|x-\mu|/b))]. \end{aligned} \tag{14}$$

For different values of μ and b, the images of CDF and PDF are shown in Fig. 4. Thus, our incentive mechanism can be translated into

$$\sum_{i=1}^{N} F(x) = \sum_{i=1}^{N} \int_{-\infty}^{m_i} f(u)du = N_0. \tag{15}$$

We can see that if $\frac{N}{2b}\int_{-\infty}^{0} esp(-\frac{|x-\mu|}{b})dx \geq N_0$, which means that more than N_0 users will report the MSP in the system. Hence, we only need to consider the case where $\frac{N}{2b}\int_{-\infty}^{0} esp(-\frac{|x-\mu|}{b})dx < N_0$. First, we have analytically investigated the scenario where $\mu > 0$. It can be noticed that each $\sum_{i=1}^{N} F(m_i;\mu,b)$ is the integral of $f(x)$ from 0 to m_0, which is the area enclosed by $f(x)$, $x = m_i$, and $y = 0$. For convenience, the area enclosed can be expressed as s_i. Then, as shown in the Fig. 5, we translate Eq.(15) into

$$s_1 + s_2 + \cdots + s_{\mathrm{n}} = N_0. \tag{16}$$

Previously, we transform the incentive problem into a nonconvex optimization problem. P is assumed to be the solution to the problem, then, the analysis is performed in the form of P.

Lemma 1: The symmetric elements in P are moved by equal distances in the same direction and the sum of the enclosed areas remains the same.

Proof: As shown in Fig. 6, it can be assumed that P contains such symmetric elements of x_1 and x_2 concerning μ. We can move x_1 or x_2 to $x = 0$ and the other to the opposite side by the same distance. Similarly, we can also move both x_1 and x_2 to $x = \mu$, as shown in Fig. 7. Then, we have

$$\begin{aligned} S_1' &= |\int_{-\infty}^{x_1} f(u)du - \int_{-\infty}^{0} f(u)du| \\ &= |\int_{-\infty}^{x_2} f(u)du - \int_{-\infty}^{0} f(u)du| \\ &= S_2', \qquad x_2 > x_1, \end{aligned} \tag{17}$$

$$\begin{aligned} S_1' &= |\int_{-\infty}^{x_1} f(u)du - \int_{-\infty}^{\mu} f(u)du| \\ &= |\int_{-\infty}^{x_2} f(u)du - \int_{-\infty}^{\mu} f(u)du| \\ &= S_2', \qquad x_2 > x_1, \end{aligned} \tag{18}$$

where S_1' and S_2' represent the change in the area after moving x_1 and x_2, respectively. Clearly, we can see that the total cost remains the same and Eq.(15) is still valid. ■

Lemma 2: No two or more elements in P are greater than 0 and less than μ.

Proof: In Fig. 8, we move x_1 and x_2 to $x = 0$ and $x = x_2'$, respectively, under the condition of $0 < x_1 < x_2 < \mu$. To keep the original area, it should satisfy $x_1 - 0 > x_2' - x_2$. However, it can be noticed that this will lead to a reduction in the total area. Hence, P is not the solution to the problem. ■

Lemma 3: All elements in P that are not less than μ are equal to each other.

Proof: As shown in Fig. 9, let us consider the case where $\mu < x_1 < x_2$. For convenience of description, we assume that there exists x' and $x_1 < x' < x_2$. Then, we move both x_1 and

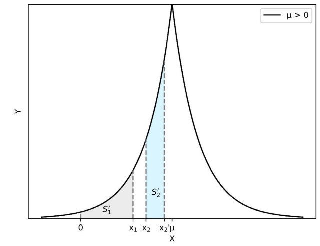

Fig. 8. $0 < x_1 < x_2 < \mu$.

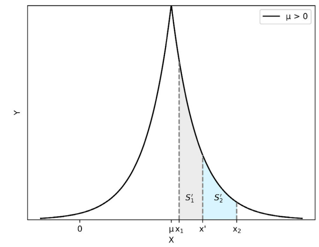

Fig. 9. $\mu < x_1 < x_2$.

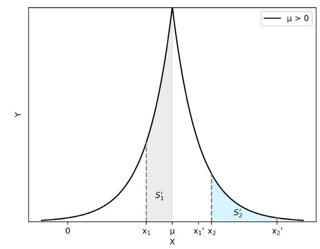

Fig. 10. $x_2 > x_1'$.

TABLE I
MALICIOUS PLATFORM DETECTION RESULTS OF PARTIAL INFORMATION SCENARIO

Scenarios	Conditions	Results
Uniform Distribution	$b<0$ or b>0, a $\leq$ 0, $N\int_a^0 \frac{1}{b-a} \geq N_0$	$m_i = 0, \forall_i$
	$b>0$, a≤0, $N\int_a^0 \frac{1}{b-a} < N_0$	$m_i = \frac{(b-a)N_0}{N} + a, \forall_i$
	$b>0,\ a>0$	$m_i = b, for\ \ i = 1, 2, \cdots N_0$; $m_i = 0, otherwise$
Laplace Distribution	$\int_{-\infty}^0 esp(-\frac{\lvert x-\mu \rvert}{b})dx \geq \frac{2bN_0}{N}$	$m_i = 0, \forall_i$
	$\int_{-\infty}^0 esp(-\frac{\lvert x-\mu \rvert}{b})dx < \frac{2bN_0}{N}$, $\mu \leq 0$	$m_i = F^{-1}(\frac{N_0}{N}), \forall_i$
	$\int_{-\infty}^0 esp(-\frac{\lvert x-\mu \rvert}{b})dx < \frac{2bN_0}{N}$, $\mu > 0$	$m_i = F^{-1}(\frac{N_0 - nF(0)}{N-n})$, $for\ \ i = 1, 2, \cdots N_0$; $m_i = 0, otherwise$

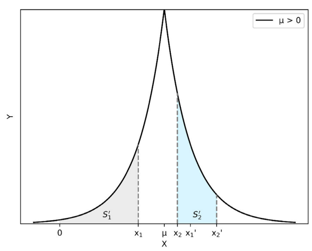

Fig. 11. $x_2 < x_1{}'$.

x_2 to $x = x'$. Similarly, the distance from x_1 to x' should be greater than the distance from x_2 to x' to keep the same area after moving x_1 and x_2. Therefore, if

$$\begin{aligned} S_1' &= \lvert \int_{-\infty}^{x'} f(u)du - \int_{-\infty}^{x_1} f(u)du \rvert \\ &= \lvert \int_{-\infty}^{x_2} f(u)du - \int_{-\infty}^{x'} f(u)du \rvert \\ &= S_2', \qquad x_2 > x_1 > u, \end{aligned} \tag{19}$$

is valid, we can get $x_1 + x_2 > 2x'$. It can be noted that the total cost will be lower and P is not the solution to the problem. ∎

Lemma 4: There are no elements in P that are greater than 0 and less than μ.

Proof: Let us consider a scenario where x_1 to x_2 are the two elements in P. According to the symmetry of the function, the symmetry point $x_1{}'$ can be found,i.e. $|x_1 - \mu| = |x_1{}' - \mu|$. Next we will analyze whether P is the solution to the problem under the three scenarios.

First, we consider the case where x_2 is greater than x_1'. We can see from this Fig. 10 that the gray area and the blue area indicate the increased area after moving x_1 to μ and the decreased area after moving x_2 to x_2', respectively. To keep the

$$\begin{aligned} S_1' &= \lvert \int_{-\infty}^{\mu} f(u)du - \int_{-\infty}^{x_1} f(u)du \rvert \\ &= \lvert \int_{-\infty}^{x_2{}'} f(u)du - \int_{-\infty}^{x_2} f(u)du \rvert \\ &= S_2', \qquad x_2 > \mu > x_1, \end{aligned} \tag{20}$$

is valid, we have $|x_1 - \mu| < |x_2 - x_2{}'|$. However, the total cost will be reduced, which lead to P is not the solution to the problem. Then, let us consider the case where x_2 is less than x_1'. The change in area after moving x_1 and x_2 is plotted in Fig. 11. Similarly, the

$$\begin{aligned} S_1' &= \lvert \int_{-\infty}^{x_1} f(u)du - \int_{-\infty}^{0} f(u)du \rvert \\ &= \lvert \int_{-\infty}^{x_2{}'} f(u)du - \int_{-\infty}^{x_2} f(u)du \rvert \\ &= S_2', \qquad x_1 < \mu < x_2, \end{aligned} \tag{21}$$

is valid on the premise that $|x_1 - 0| < |x_2 - x_2{}'|$. It can be noted that the total cost will decrease as the elements move. Hence, P is not the solution to the problem. At last we investigate the case where $x_1' = x_2$. It is clear that this case is the same as $Proof$4.1, the total area of the elements symmetric about μ remains the same after moving to 0 or μ. However, this leads to inequality between elements larger than μ. Therefore, P is not the solution to the problem. ∎

According to the previous proof, we can find that the optimal solution can be obtained only when some elements are 0 and others are greater than or equal to μ. In other words, there exist n elements equal to 0 and all elements greater than μ are m. Therefore, we have

$$(N-n)F(m) + nF(0) = N_0, \tag{22}$$

then

$$m = F^{-1}(\frac{N_0 - nF(0)}{N-n}). \tag{23}$$

Thus from (22) and (23), the cost total can be obtained

$$M = (N-n)F^{-1}(\frac{N_0 - nF(0)}{N-n}). \tag{24}$$

Based on the fact that $0 \leq n < N$ and n is an integer, the above incentive design problem can be transformed into computing the n. The solution process of n is described in Algorithm 1. After the value of n is determined, the system provides incentives to n and $N-n$ users in the system as 0 and $F^{-1}\frac{N_0 - nF(0)}{N-n}$, respectively.

Next, we analyze the two cases where $\mu < 0$ and $\mu = 0$. Similar to *lamma 4.3*, we can conclude that all elements are the same and no element is equal to 0. Therefore, the incentive scheme is to give each user the same m,

$$m = F^{-1}(\frac{N_0}{N}). \tag{25}$$

Algorithm 1 Laplace Cost Optimization

Require: N;N_0;
Ensure: M,n

$n \longleftarrow 1$;
$M \longleftarrow NF^{-1}\frac{N_0-nF(0)}{N-n}$;
while $n \leq N$ **do**
  $M' \longleftarrow (N-n)F^{-1}\frac{N_0-nF(0)}{N-n}$;
  if $M' < M$ **then**
    $M \longleftarrow M'$;
    $n \longleftarrow n+1$;
  end if
end while
return (M, n)

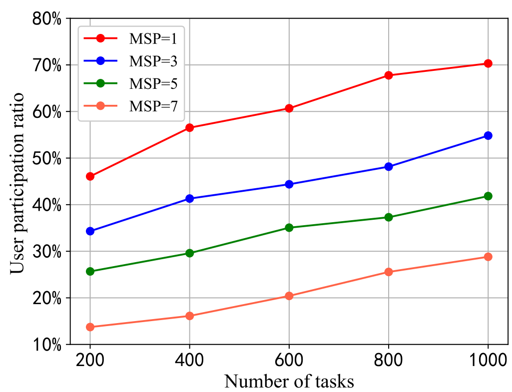

Fig. 12. The participation ratio with different number of MSP and task.

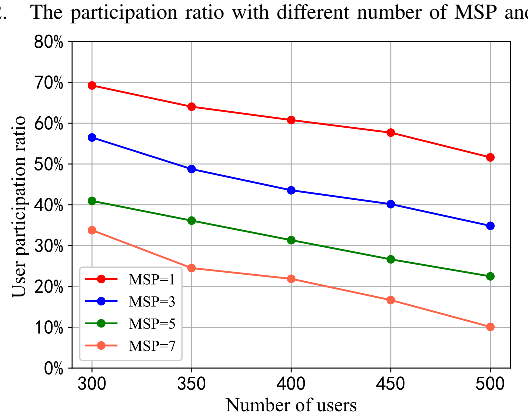

Fig. 13. The participation ratio with different number of MSP and user.

The results of the partial information scenario are shown in Table I.

V. Performance Evaluation

In this section, we evaluate the performance of the proposed algorithm and the impact of different parameters on the total cost through simulation.

A. Impact of MSP

We first evaluate the impact of the number of MSP on the participation ratio, where the number of sensing platform is 10. With the increased number of tasks when the number of users is 200, the impact of the number of MSP on the participation ratio is shown in Fig. 12. We can see from this figure that the participation ratio of four scenarios increases as the number of tasks increases. This is because when the number of tasks is relatively less than the number of participants, the number of participants who can undertake the task is also relatively small. Therefore, the participation ratio of all three scenarios is at a low level. As shown in Fig. 13, the participation ratio of the four scenarios decreases as the number of users increases when the number of tasks is 1000. This is because the increasing number of users quickly saturates the number of tasks that can be performed.

From the result of Fig. 12 and Fig. 13, it can be noted that in the same condition, the higher the number of MSP the lower the participation ratio. The reason is that users may fear that their interests are being violated by malicious platforms, therefore, they refuse to join perception activities.

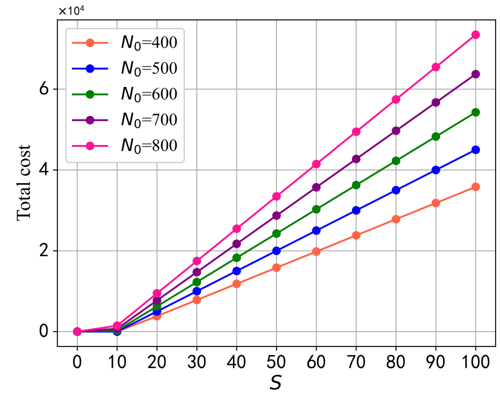

Fig. 14. Impact of S and N_0 on the total cost.

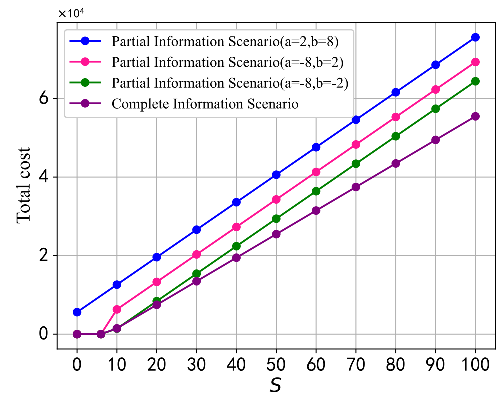

Fig. 15. Impact of S on the total cost in the uniform distribution.

B. Complete Information Scenario

We assume that the user's preference p_i follows the Laplace Distribution $p_i \sim$ *Laplace(-10,2)* in the complete information scenario. Then, the system evaluates the impact on the total cost by changing the motivation of the MSP. To observe the total cost, different values of S (from 0 to 100) are set previously. As shown in Fig. 14, the total cost tends to increase as the number of N_0. It can be observed that the total cost for different numbers of N_0 on the left side is 0. The reason is that the sum of the user's preference and the MSP's motivation is less than 0, i.e., $p_{N_i} \leq -s$, which means that more than N_0 users will report MSP. Therefore, the system does not need to provide additional rewards. Simultaneously, as the motivation S increases, we can see that the total cost will also increase. This is due to the fact that the increase of S will lead

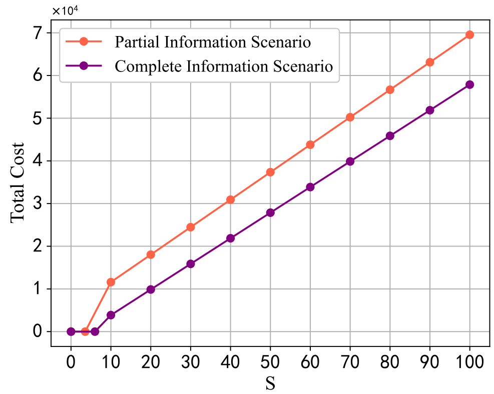

Fig. 16. Impact of S on the total cost in the laplace distribution.

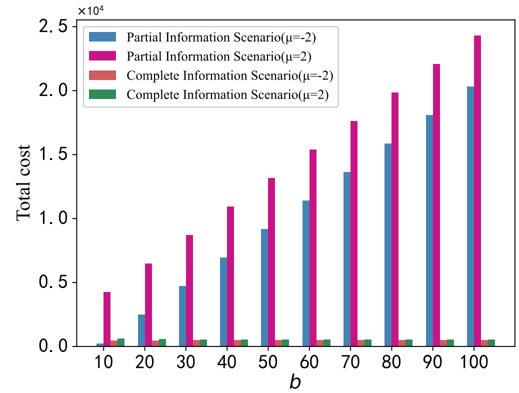

Fig. 17. Impact of b on the total cost in the laplace distribution.

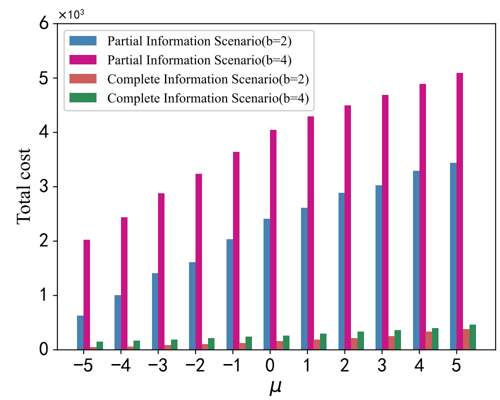

Fig. 18. Impact of μ on total cost in the laplace distribution.

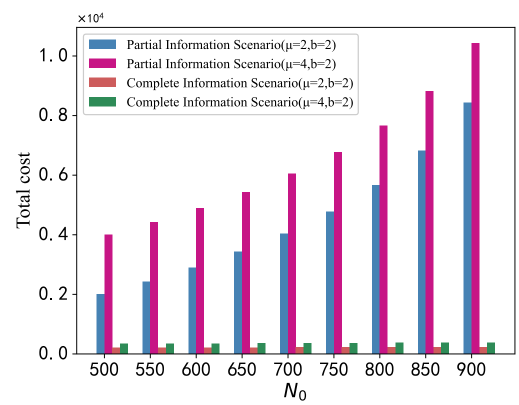

Fig. 19. Impact of N_0 on total cost in the laplace distribution.

to the decrease in the number of users whose preferences are not higher than $-S$.

C. Uniform Distribution Scenario

In the case, where $N = 1000$ and $N_0 = 600$, we set the user's preferences to obey $p_i \sim$ *Uniform(2,8)*, $p_i \sim$ *Uniform (-8,2)*, and $p_i \sim$ *Uniform(-8,-2)* respectively. As shown in Fig. 15, we compare the impact of S on total cost in the partial information scenario and the complete information scenario. It can be observed that the total cost increases with the increase of S. The cost of the three cases $p_i \sim$ *Uniform(-8,2)*, $p_i \sim$ *Uniform(-8,-2)* and the complete information scenario is 0, when the S is relatively small. The reason is that most users will choose to report MSP. The slope of $p_i \sim$ *Uniform (-8,2)* and $p_i \sim$ *Uniform(-8,-2)* will change from N to N_0 when $p_{N_i} + S > 0$. This is because the total cost is determined by Eq.(11) in the initial stage, but as S increases the total cost will be determined by $N_0 b$.

Furthermore, Fig. 15 demonstrates that the total cost of $p_i \sim$ *Uniform(2,8)* is higher than that of $p_i \sim$ *Uniform(-8,-2)* and $p_i \sim$ *Uniform(-8,2)* as S increases. Since, under the conditions of $p_i \sim$ *Uniform(2,8)*, the user's preferences for MSP are non-negative, the system needs to provide more incentives for users to report MSP. This also proves that the cost of $p_i \sim$ *Uniform(2,8)* is not 0 when S is small. However, in the real world, $p_i \sim$ *Uniform(2,8)* is non-existent. Because users have a negative preference for malicious platforms in most cases. Users of $p_i \sim$ *Uniform(-8,2)* are more tolerant than users of $p_i \sim$ *Uniform(-8,-2)* towards MSP, so the total cost of the former is greater than the latter.

D. Laplace Distribution Scenario

Similarly, we set the user's preference to match $p_i \sim$ *Laplace(-8,2)* and the threshold is $N_0 = 600$. Then, the change of total cost in the partial information scenario and the full information scenario is observed by changing S of MSP, which is plotted in Fig. 16. It can be noted that the total cost of the two scenarios increases as the S increases. This is because the total cost in both scenarios is a linear function of S. Furthermore, when n is determined, the effect of $F(0)$ is ignorable. Intuitively, the total cost in the full information scenario is lower than that in the partial information scenario under a certain S.

In the following, we observe the total cost by changing the value of b. In the Fig. 17, it can be observed that the total cost of the partial information scenario shows an upward trend as the b increase, where $N_0 = 600$, $p_i \sim$ *Laplace(-2,b)* and $p_i \sim$ *Laplace(2,b)*. The reason for this is that the total cost of the partial information scenario is determined by $(N-n)F^{-1}(\frac{N_0-nF(0)}{N-n})$ and $NF^{-1}(\frac{N_0}{N})$. When n is given by Algorithm 1, the total cost and b are linearly related. But, we can see that the total cost of the full information scenario does not increase rapidly with b, which means that it's performance is better than the partial information scenario.

Simultaneously, the impact of the value of μ on the total cost is plotted in Fig. 18, where $N_0 = 600$. In Fig. 18, it can be seen that as the value of μ increases, the total cost also increases. The reason for this is that an increase in μ means

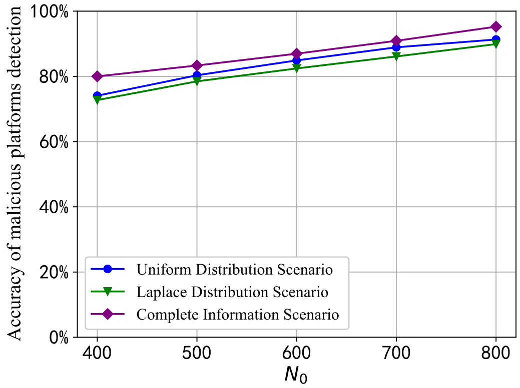

Fig. 20. Comparison of accuracy.

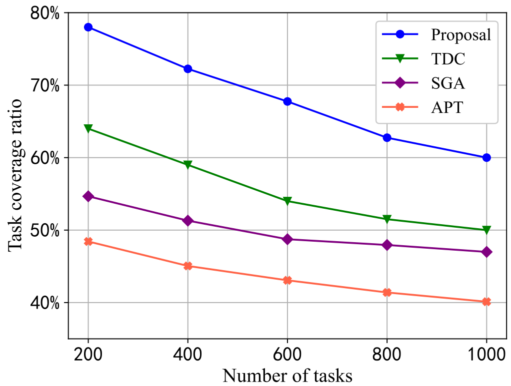

Fig. 21. Comparison of task coverage ratio.

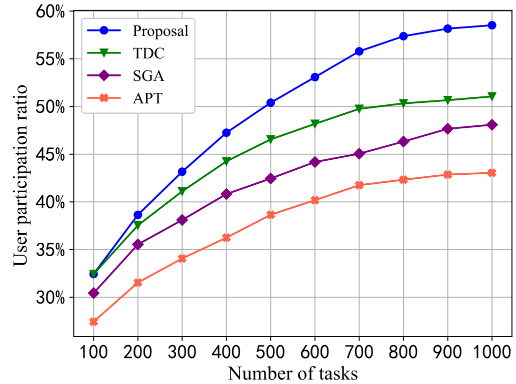

Fig. 22. Impact of different number of tasks on participation ratio.

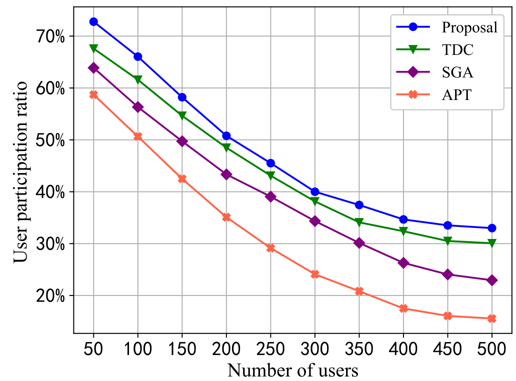

Fig. 23. Impact of different number of users on participation ratio.

that the number of users with a low preference for MSP is decreasing. Thus, the system must pay more rewards to motivate users to report MSP. The total cost is determined by $NF^{-1}(\frac{N_0}{N})$ and $(N-n)F^{-1}(\frac{N_0-nF(0)}{N-n})$ for conditions $\mu \leq 0$ and $\mu > 0$, respectively. This leads to the fact that the total cost in the partial information scenario increases at a different rate in the two cases $\mu \leq 0$ and $\mu > 0$.

To make the experimental results more rigorous, we investigate the change of the total cost under different N_0, where p_i $\sim$ *Laplace(2,2)* and $p_i \sim$ *Laplace(4,2)*. As shown in Fig. 19, in the partial information scenario, the total cost shows an upward trend as N_0 increases. Simultaneously, we can note that the total cost of $\mu = 2$ is lower than the total cost of $\mu = 4$ when the N_0 is certain. This is because the value of $(N-n)F^{-1}(\frac{N_0-nF(0)}{N-n})$ increases as μ increases when b is certain, which makes the total cost increase. Moreover, in the full information scenario, the total cost increases slowly as N_0 increases.

E. Accuracy of MSP Detection

We assume separately that the user's preferences obey p_i $\sim$ *Uniform(-8,2)* and $p_i \sim$ *Laplace(4,2)*. Next, we analyzed the accuracy of MSPs detection under different N_0, where $S = 60$ and $N = 1000$. From Fig. 20, it can be seen that all three scenarios show an increasing trend as N_0 increases. This is because the increase in N_0 reduces the impact of some users reporting NSPs, which makes detection more accurate. In complete information scenario, the system is fully aware of the user's preferences, which means that most users choose to report MSPs. Therefore, the accuracy of complete information scenario is higher than uniform distribution scenario and laplace distribution scenario.

F. Task Coverage Ratio

We assume that the user's preferences obey $p_i \sim$ *Laplace(-8,2)* and $N_0 = 120$. Then, five MSPs are put into the sensing system and release sensing tasks with the other five NSPs. Fig. 21 shows the comparison of the task coverage ratio by TDC(A Trust-Driven Contract Incentive Scheme for Mobile Crowd-Sensing Networks) [37], SGA(A Stackelberg Game Approach Toward Socially-Aware Incentive Mechanisms for Mobile Crowdsensing) [39], APT(Accurate and Privacy-Preserving Task Allocation for Edge Computing Assisted Mobile Crowdsensing) [40] and our algorithm. Under the number of users is 200, we can observe that all four algorithms have different degrees of decline as the number of tasks increases. This is because when the number of participants is fixed, the number of participants who can perform tasks will reach saturation. It can be noticed that the task coverage ratio of TDC is higher than SGA and APT, but lower than our algorithm. The reason for this is that the TDC filters MSPs by trust thresholds, however, the MSP can provide motivation to trick user's confidence. The reason for the lowest task coverage ratio of SGA and APT is that there is no mechanism to remove MSPs in SGA. SGA encourages users to complete sensing tasks by offering them rewards, but when the MSPs exist, the rewards are not valid for users. APT focus on task allocation and users' privacy protection, therefore, the existence of MSPs has the greatest impact on it's task coverage ratio. To sum up, our algorithm is superior to the other three algorithms.

G. User Participation Ratio

Finally, the participation ratios of we proposed and the other three mechanisms are compared. The user's preferences obey $p_i \sim$ *Laplace(-8,2)* and $N_0 = 120$. Then, we set five MSPs and five MNPs in the simulation environment. With the number of tasks increasing, the evaluation results on the five mechanisms are plotted in Fig. 22, where the number of users is 200. We can observe that the participation ratio of the four mechanisms increases as the number of tasks increases. The reason for this is that when the number of tasks is relatively less than the number of participants, the number of participants who can undertake the task is also relatively small. However, as the number of tasks increases, the participation ratio of the four mechanisms shows an increasing trend.

As shown in Fig. 23, the participation ratio of these mechanisms decreases as the number of participants increases, where the number of tasks is 1000. This is because that the increasing number of participants quickly saturates the number of tasks, which can be performed when the number of tasks is fixed. Thus when the number of remaining tasks is insufficient, many participants are unable to undertake tasks. It can be noted that the mechanism of we proposed is superior to the other three mechanisms. Because APT focuses on user privacy, it has the lowest performance when MSPs are present in the system.

VI. Conclusion

In this paper, we have analytically investigated the MSP detection problem in MCS. To encourage users to participate in sensing activities, we design the MSP detecting algorithm in terms of user's preferences. Two scenarios, namely complete information scenario and partial information scenario, have been considered to detect the MSP when users join sensing activities. In the complete information scenario, the system is fully aware of the user's preferences. Therefore, we design an incentive mechanism that makes users report MSPs by ranking their preferences. We assume that the users' preferences obey Uniformly Distributed and Laplace Distributed in the complete information scenario. Then the incentive mechanism for them is designed, respectively, which to optimize the total incentive cost. These incentive mechanisms help us to understand the impact of different user's preferences on the detection of MSPs, and the corresponding analytical results can serve as important guidelines on the design of more sophisticated detection schemes of MSPs. The soundness of the models and the accuracy of the analysis have been verified through extensive simulation. Our simulation results have shown that the impact of different factors on total cost and the proposed incentive mechanism outperforms other benchmarks substantially.

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